

Bandwidth Utilization



Course Code: COE 3201

Course Title: Data Communication

**Dept. of Computer Engineering
Faculty of Engineering**

Lecture No:	9	Week No:	11

Lecture Outline



- ☐ Wavelength-division multiplexing (WDM)
- ☐ Time-division multiplexing (TDM)
- ☐ Synchronous TDM
- ☐ Interleaving
- ☐ Data rate management
- ☐ Digital Hierarchy
- ☐ T-lines & E-Lines

Wavelength-division multiplexing (WDM)



- ❑ Wavelength-division multiplexing (WDM) is designed to use the high-data-rate capability of fiber-optic cable.
- ❑ The optical fiber data rate is higher than the data rate of metallic transmission cable but using a fiber-optic cable for a single line wastes the available bandwidth.
- ❑ WDM allows us to combine several lines into one.

- ❑ WDM is conceptually the same as FDM, except that the multiplexing and demultiplexing involve optical signals transmitted through fiber-optic channels.
- ❑ The idea is the same: We are combining different signals of different frequencies.
- ❑ The difference is that the frequencies are very high.
- ❑ Figure 6.10 gives a conceptual view of a WDM multiplexer and demultiplexer.
- ❑ Very narrow bands of light from different sources are combined to make a wider band of light. At the receiver, the signals are separated by the demultiplexer.

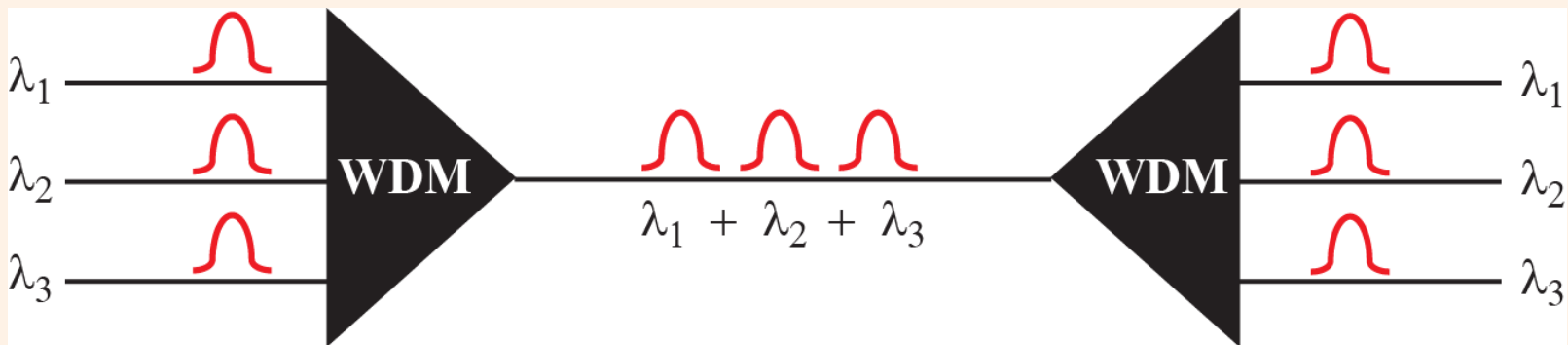


Figure 6.10: Wavelength-division multiplexing

- ❑ Although WDM technology is very complex, the basic idea is very simple.
- ❑ We want to combine multiple light sources into one single light at the multiplexer and do the reverse at the demultiplexer.
- ❑ The combining and splitting of light sources are easily handled by a prism.

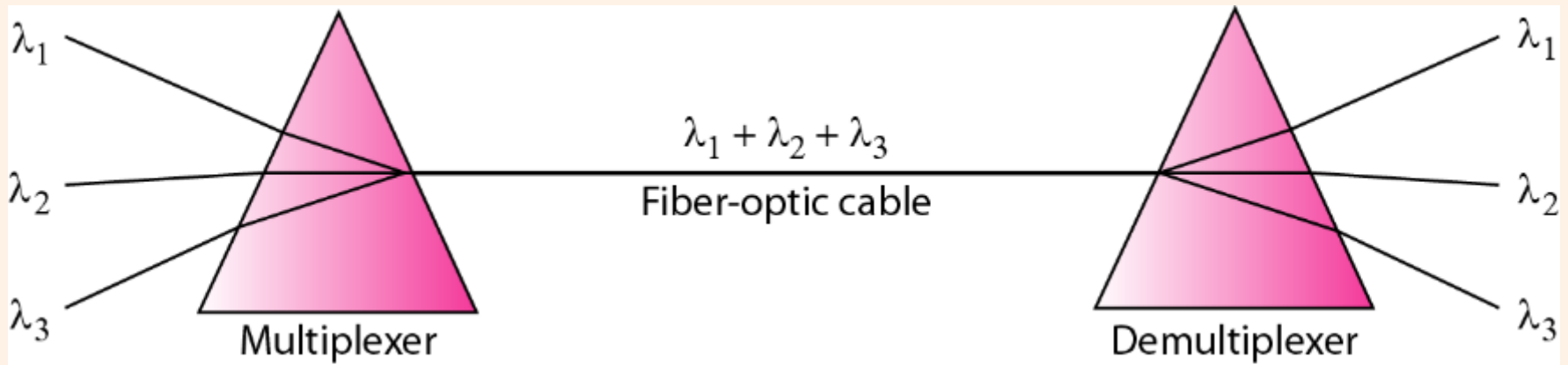


Figure 6.11: Prisms in wave-length division multiplexing

Time-division Multiplexing (TDM)



- ❑ Time-division multiplexing (TDM) is a digital process that allows several connections to share the high bandwidth of a link.
- ❑ Instead of sharing a portion of the bandwidth as in FDM, time is shared. Each connection occupies a portion of time in the link.
- ❑ Figure 6.12 gives a conceptual view of TDM.
- ❑ Note that the same link is used as in FDM; here, however, the link is shown sectioned by time rather than by frequency.

In the figure, portions of signals 1, 2, 3, and 4 occupy the link sequentially.

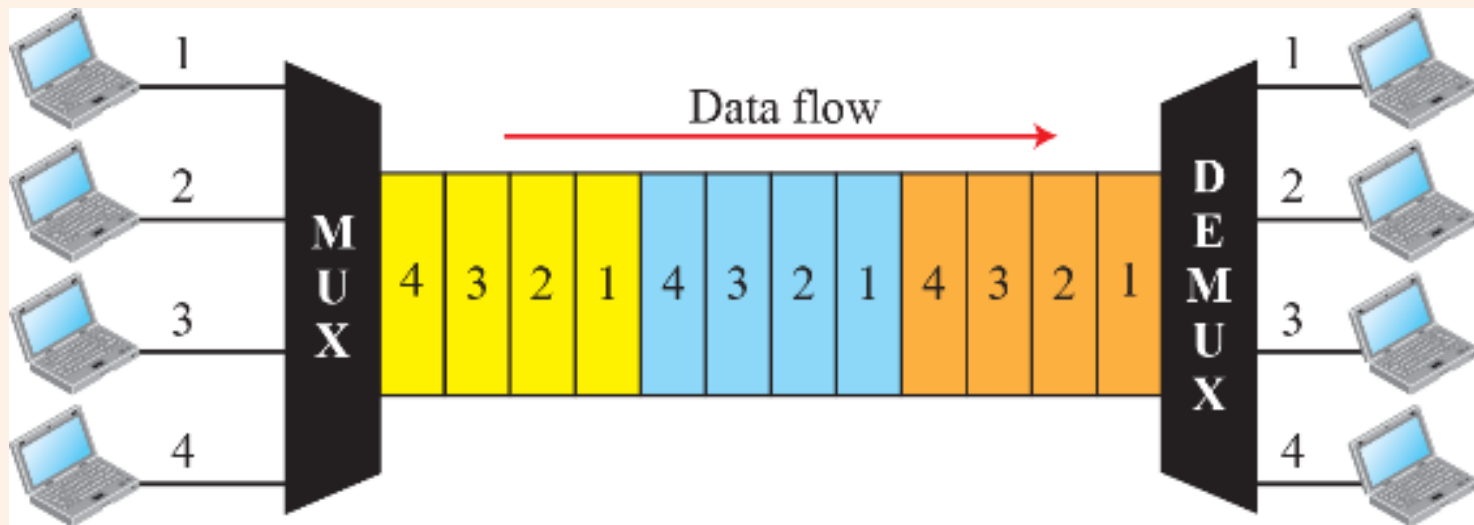


Figure 6.12: TDM

Synchronous TDM

- ❑ We can divide TDM into two different schemes: **synchronous** and **statistical**.
- ❑ In synchronous TDM, each input connection has an allotment in the output even if it is not sending data.
- ❑ Time Slots and Frames: In synchronous TDM, the data flow of each input connection is divided into units, where each input occupies one input time slot. Each input unit becomes one output unit and occupies one output time slot. However, the duration of an output time slot is n times shorter than the duration of an input time slot.

Synchronous TDM

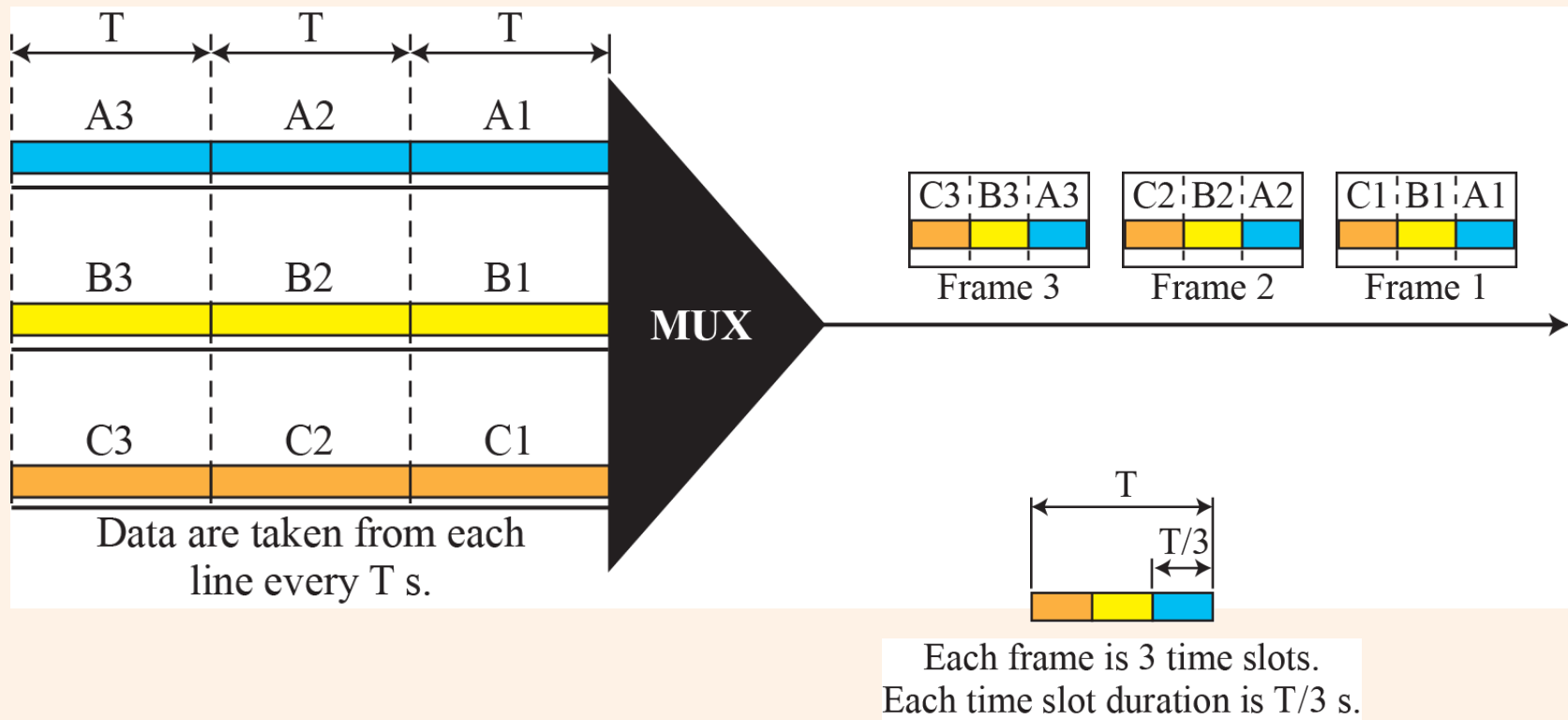


Figure 6.13: Synchronous time-division multiplexing

Example 6.5: In Figure 6.13, the data rate for each input connection is 1 kbps. If 1 bit at a time is multiplexed (a unit is 1 bit), what is the duration of a. each input slot, b. each output slot, and c. each frame?

Solution: We can answer the questions as follows:

- a. The data rate of each input connection is 1 kbps. This means that the bit duration is $1/1000$ s or 1 ms. The duration of the input time slot is 1 ms (same as bit duration).
- b. The duration of each output time slot is one-third of the input time slot. This means that the duration of the output time slot is $1/3$ ms.
- c. Each frame carries three output time slots. So the duration of a frame is $3 \times (1/3)$ ms, or 1 ms. The duration of a frame is the same as the duration of an input unit.

Example 6.6: Figure 6.14 shows synchronous TDM with a data stream for each input and one data stream for the output. The unit of data is 1 bit. Find (a) the input bit duration, (b) the output bit duration, (c) the output bit rate, and (d) the output frame rate.

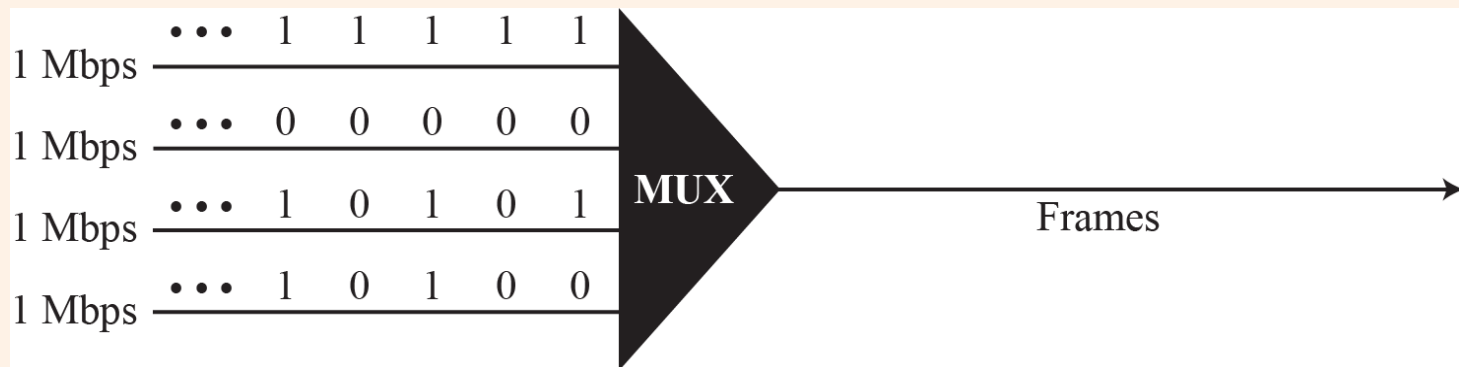


Figure 6.14: Example 6.6

Synchronous TDM



Solution

We can answer the questions as follows:

1. The input bit duration is the inverse of the bit rate: $1/1 \text{ Mbps} = 1 \mu\text{s}$.
2. The output bit duration is one-fourth of the input bit duration, or $1/4 \mu\text{s}$.
3. The output bit rate is the inverse of the output bit duration, or $1/(1/4) \mu\text{s}$ or 4 Mbps. This can also be deduced from the fact that the output rate is 4 times as fast as any input rate; so the output rate $= 4 \times 1 \text{ Mbps} = 4 \text{ Mbps}$.
4. The frame rate is always the same as any input rate. So the frame rate is 1,000,000 frames per second. Because we are sending 4 bits in each frame, we can verify the result of the previous question by multiplying the frame rate by the number of bits per frame.

Example 6.7: Four 1-kbps connections are multiplexed together. A unit is 1 bit. Find (1) the duration of 1 bit before multiplexing, (2) the transmission rate of the link, (3) the duration of a time slot, and (4) the duration of a frame.

Solution

We can answer the questions as follows:

1. The duration of 1 bit before multiplexing is $1/1$ kbps, or 0.001 s (1 ms).
2. The rate of the link is 4 times the rate of a connection, or 4 kbps.
3. The duration of each time slot is one-fourth of the duration of each bit before multiplexing, or $1/4$ ms or 250 μ s. Note that we can also calculate this from the data rate of the link, 4 kbps. The bit duration is the inverse of the data rate, or $1/4$ kbps or 250 μ s.
4. The duration of a frame is always the same as the duration of a unit before multiplexing, or 1 ms. We can also calculate this in another way. Each frame in this case has four time slots. So the duration of a frame is 4 times 250 μ s, or 1 ms.

Interleaving

The **process** of taking a **group of bits** from **each input line** for multiplexing is called ***interleaving***.

Interleaving is the process of **multiplexing**.

In **TDM**, **synchronization** between the **sender** and **receiver** is very important.

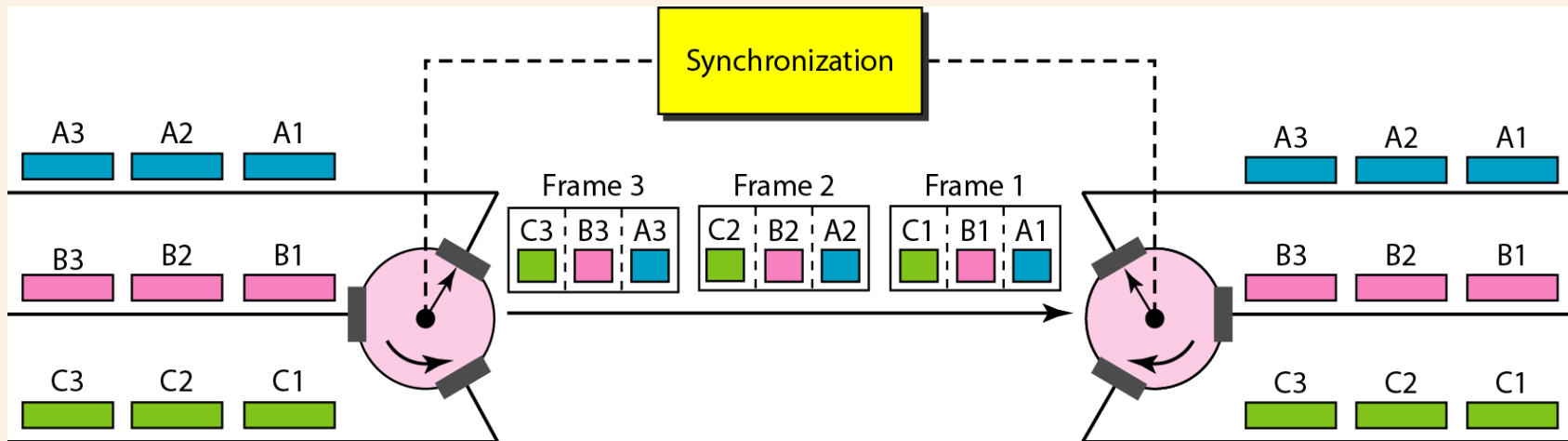
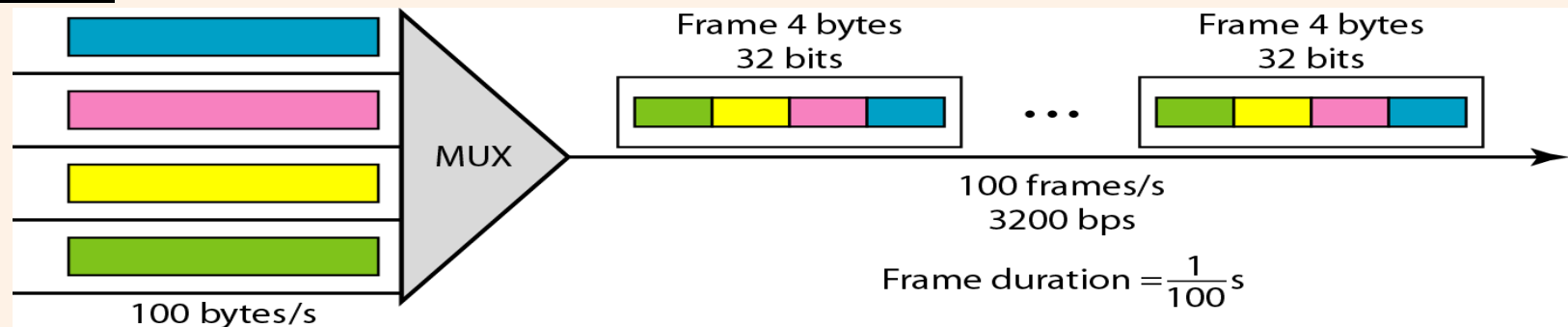


Fig. 17: Interleaving

TDM

Example 6.8: Four channels are multiplexed using TDM. If each channel sends **100 bytes /s** and we multiplex **1 byte per channel**, show the **frame traveling on the link**, the **size of the frame**, the **duration of a frame**, the **frame rate**, and the **bit rate for the link**.

Solution:



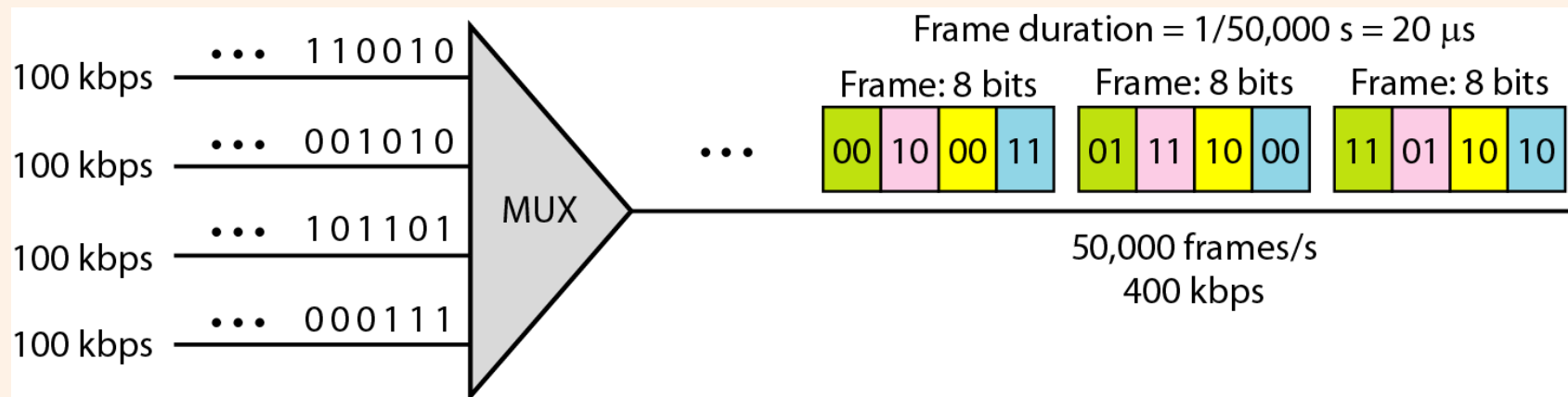
The multiplexer is shown in **Figure 18**. Each frame carries **1 byte** from each channel; the size of each frame, therefore, is **4 bytes, or 32 bits**. Because each channel is sending **100 bytes/s** and a frame carries **1 byte** from each channel, the frame rate must be **100 frames per second**. The bit rate is **100 × 32, or 3200 bps**.

Example 6.9:

A multiplexer combines **four 100-kbps channels** using a time slot of **2 bits**. Show **the output with four arbitrary inputs**. What is the **frame rate**? What is the **frame duration**? What is the **bit rate**? What is the **bit duration**?

Solution:

Figure 19 shows the output (**4x100kbps**) for **four arbitrary inputs**. The link carries **$400K/(2 \times 4) = 50,000$** frames per second [**$2 \times 4 = 8\text{bit}$**]. The frame duration is therefore **$1/50,000$ s** or **$20 \mu\text{s}$** . The **bit duration** on the output link is **$1/400,000$ s**, or **$2.5 \mu\text{s}$** .



Empty Slot

- ❑ Sometimes an input link may **have no data to transmit**.
- ❑ When that happens, **one or more slots** on the output link will go **unused**.
- ❑ That is **wastage of bandwidth**.
- ❑ **Statistical TDM** can improve the efficiency.

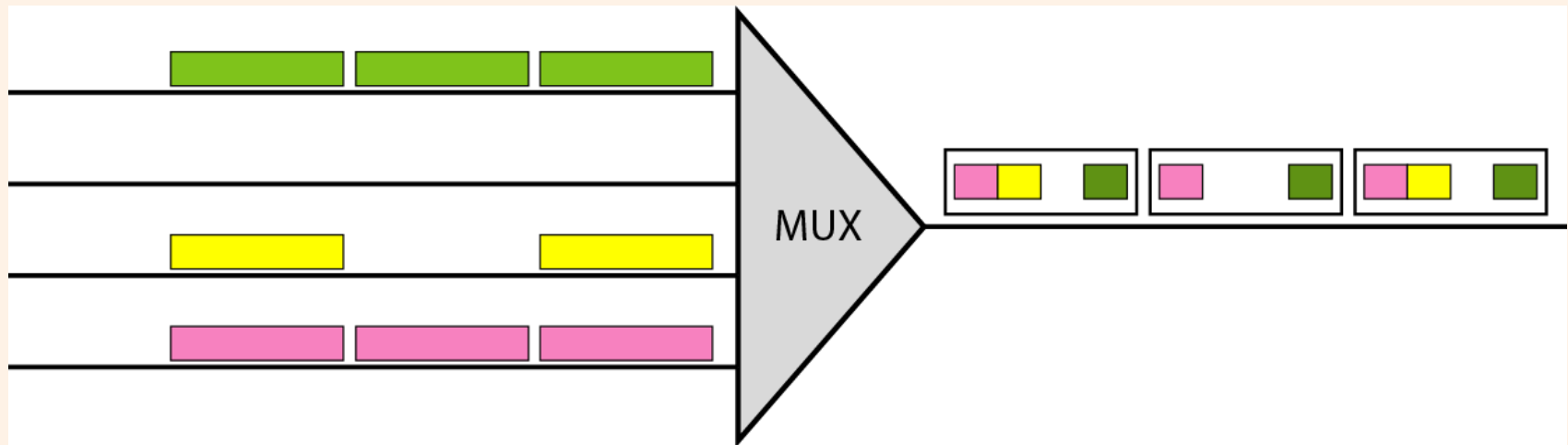


Fig. 20: Empty Slots.

Data Rate Management

❑ Data Rate Management:

Not all input links maybe have the **same data rate**.

Some links maybe **slower**. There maybe several different input link speeds.

There are **three strategies** that can be used to overcome the data rate mismatch: ***multilevel, multislot and pulse stuffing.***

❑ Data rate matching:

Multilevel: used when the **data rate of the input links** are **multiples** of each other.

Multislot: used when there is a **GCD(Greatest Common Divisor)** between the data rates. The **higher bit rate channels** are **allocated** more slots per frame, and the **output frame rate** is a multiple of each input link.

Pulse Stuffing: used when there is **no GCD between the links**. The **slowest** speed link will be **brought up to the speed** of the other links by **bit insertion**, this is called **pulse stuffing**.

Data Rate Management

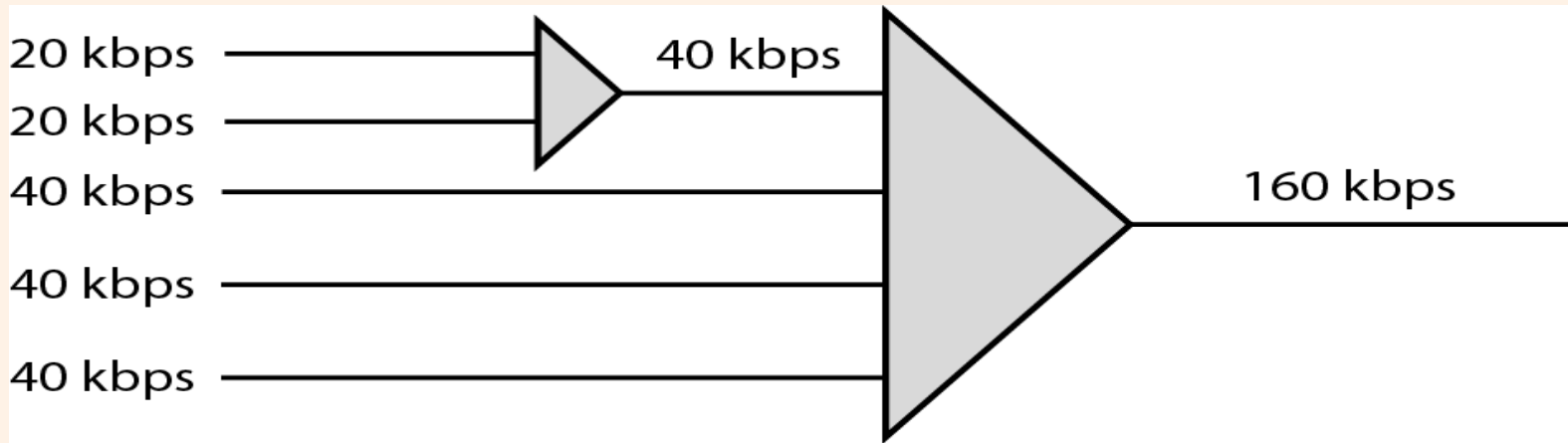


Fig. 21: Multilevel Multiplexing.

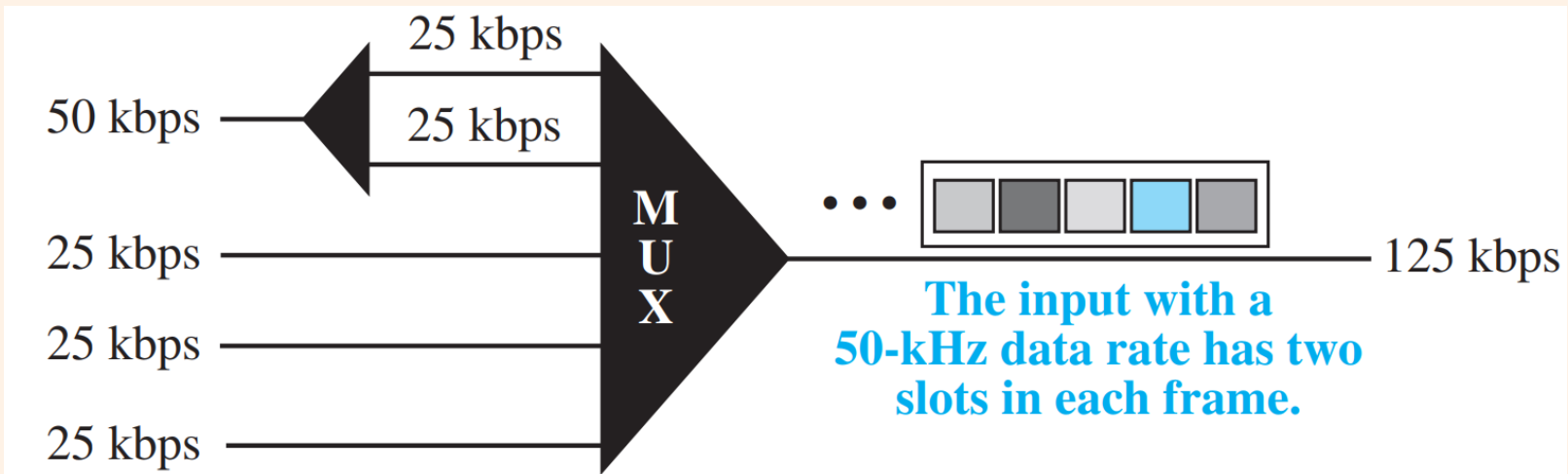


Fig. 22: Multislot Multiplexing.

Data Rate Management

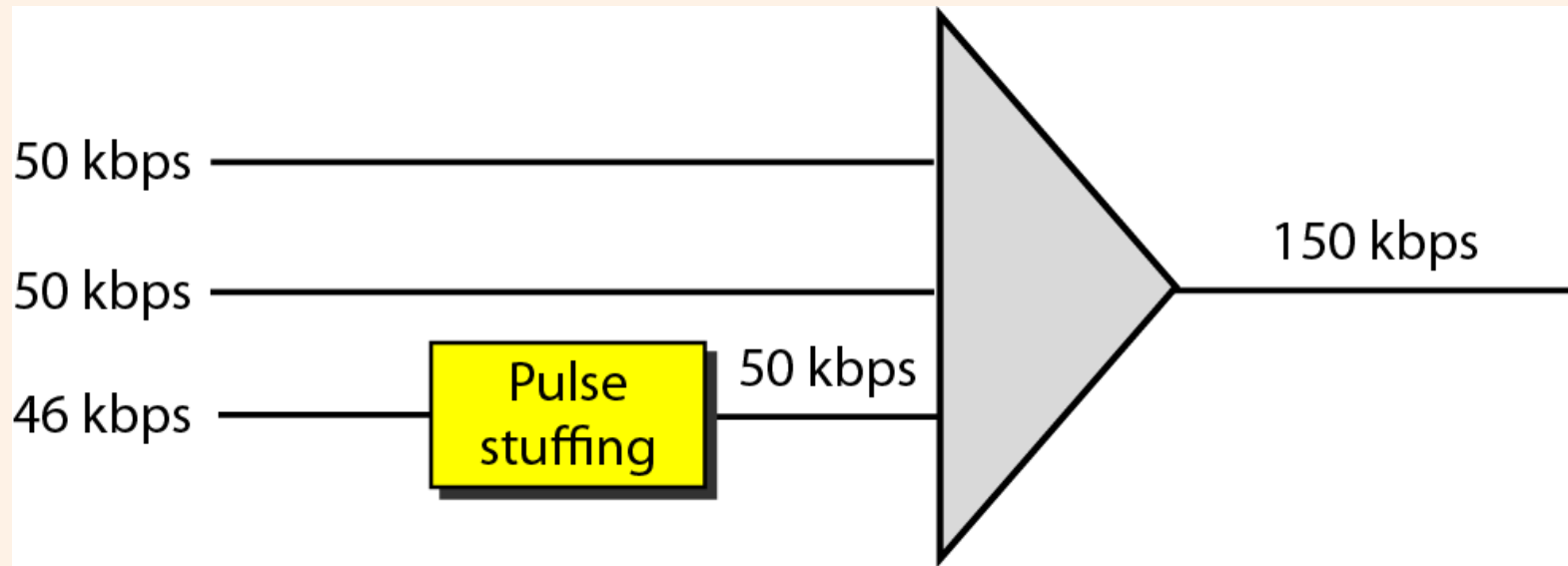


Fig. 23: Pulse stuffing Multiplexing

Frame Synchronization

- ❑ Synchronization between the **multiplexer** and **demultiplexer** is a major issue.
- ❑ If the **multiplexer** and the **demultiplexer** are **not synchronized**, a **bit belonging to one channel** may be **received by the wrong channel**.
- ❑ For this reason, **one or more synchronization bits** are usually **added** to the **beginning** of each frame.
- ❑ These bits, called **framing bits**, follow a **pattern**, frame to frame, that **allows** the **demultiplexer to synchronize** with the **incoming stream** so that it can **separate** the time slots **accurately**.
- ❑ In most cases, this synchronization information consists of **1 bit per frame**, alternating between **0** and **1**, as shown in **Figure 24**.

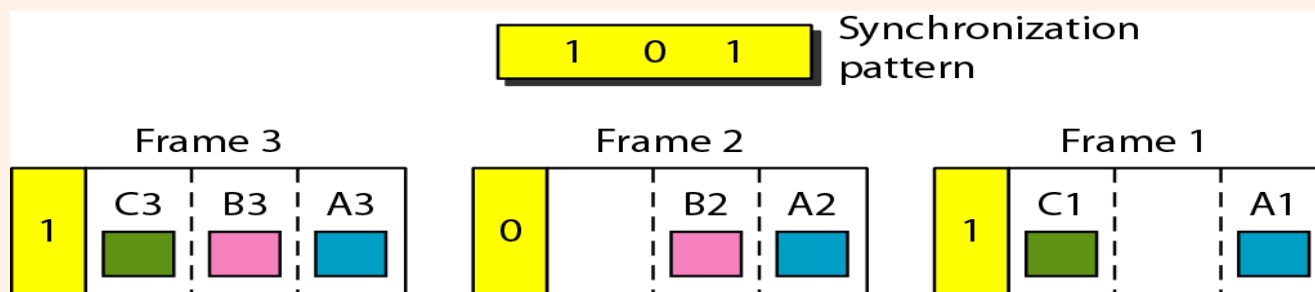


Fig. 24: Framing bits.

Mathematical Problem

Example 6.10:

We have **four sources**, each creating **250 8-bit** characters per second. If the **interleaved** unit is a **character** and **1 synchronizing bit** is **added** to each frame, find (a) the **data rate of each source**, (b) the **duration of each character in each source**, (c) the **frame rate**, (d) the **duration of each frame**, (e) the **number of bits in each frame**, and (f) the **data rate of the link**.

Solution:

- a.** The **data rate** of each source is $250 \times 8 = 2000 \text{ bps} = 2 \text{ kbps}$.
- b.** Each source sends **250 characters per second**; therefore, the **duration** of a character is $1/250 \text{ s}$, or **4 ms**.
- c.** Each **frame** has **one character** from each source, which means the link needs to send **250 frames per second** to keep the transmission rate of each source.
- d.** The **duration** of each frame is $1/250 \text{ s}$, or **4 ms**. Note that the **duration** of each frame is the **same** as the duration of **each character** coming from each source.
- e.** Each frame **carries 4 characters** and **1 extra synchronizing bit**. This means that each frame is $4 \times 8 + 1 = 33 \text{ bits}$.
- f.** The link sends 250 frames per second, and each frame contains 33 bits. This means that the data rate of the link is 250×33 , or **8250 bps**. Note that the bit rate of the link is greater than the combined bit rates of the four channels. If we add the bit rates of four channels, we get **8000 bps**. Because 250 frames are traveling per second and each contains **1 extra bit for synchronizing**, we need to add 250 to the sum to get 8250 bps.

Mathematical Problem

Example 6.11:

Two channels, one with a **bit rate** of **100 kbps** and another with a **bit rate** of **200 kbps**, are to be multiplexed. **How to solve this data rate mismatch problem? What is the frame rate? What is the frame duration? What is the bit rate of the link?**

Solution:

We can **allocate** one slot to the **first channel** and **two slots** to the **second channel**. Each frame carries **3 bits**. The **frame rate** is **100,000 frames per second** because it carries **1 bit** from the first channel. The **bit rate** is **100,000 frames/s × 3 bits per frame, or 300 kbps**.

Note: Because each frame carries 1 bit from the first channel, the bit rate for the first channel is preserved. The bit rate for the second channel is also preserved because each frame carries 2 bits from the second channel.

Digital Hierarchy

Digital hierarchy:

Telephone companies implement TDM through a hierarchy of digital signals, called digital signal (DS) service or digital hierarchy. **Figure 25** shows the data rates supported by each level.

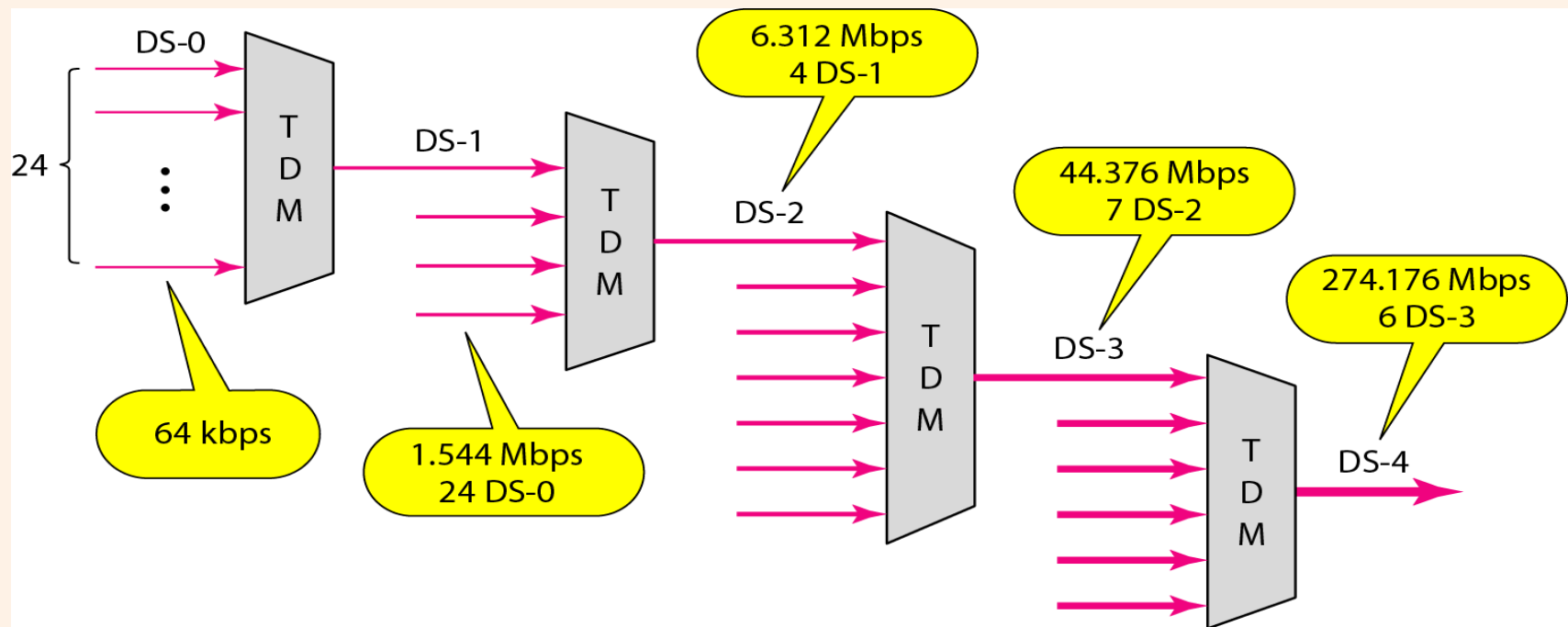


Fig. 25: Digital Hierarchy.

Digital Hierarchy

- ❑ **DS-0** is a single digital channel of 64 kbps.
- ❑ **DS-1** is a 1.544-Mbps service; 1.544 Mbps is 24 times 64 kbps plus 8 kbps of overhead. It can be used as a single service for 1.544-Mbps transmissions, or it can be used to multiplex 24 DS-0 channels or to carry any other combination desired by the user that can fit within its 1.544-Mbps capacity.
- ❑ **DS-2** is a 6.312-Mbps service; 6.312 Mbps is 96 times 64 kbps plus 168 kbps of overhead. It can be used as a single service for 6.312-Mbps transmissions; or it can be used to multiplex 4 DS-1 channels, 96 DS-0 channels, or a combination of these service types.
- ❑ **DS-3** is a 44.376-Mbps service; 44.376 Mbps is 672 times 64 kbps plus 1.368 Mbps of overhead. It can be used as a single service for 44.376-Mbps transmissions; or it can be used to multiplex 7 DS-2 channels, 28 DS-1 channels, 672 DS-0 channels, or a combination of these service types.
- ❑ **DS-4** is a 274.176-Mbps service; 274.176 is 4032 times 64 kbps plus 16.128 Mbps of overhead. It can be used to multiplex 6 DS-3 channels, 42 DS-2 channels, 168 DS-1 channels, 4032 DS-0 channels, or a combination of these service types.

Digital Hierarchy

T Lines:

DS-0, DS-1, and so on are the ***names of services***. To implement those services, the telephone companies use ***T lines (T-1 to T-4)***. These are lines with capacities precisely matched to the data rates of the ***DS-1 to DS-4*** services (see ***Table 1***). So far only ***T-1 and T-3*** lines are commercially available.

<i>Service</i>	<i>Line</i>	<i>Rate (Mbps)</i>	<i>Voice Channels</i>
DS-1	T-1	1.544	24
DS-2	T-2	6.312	96
DS-3	T-3	44.736	672
DS-4	T-4	274.176	4032

Table 1

Digital Hierarchy

T Lines for Analog Transmission:

T lines are digital lines designed for the transmission of digital data, audio, or video. However, they also can be **used for analog transmission (regular telephone connections)**, provided the **analog signals** are first **sampled**, then **time-division multiplexed**.

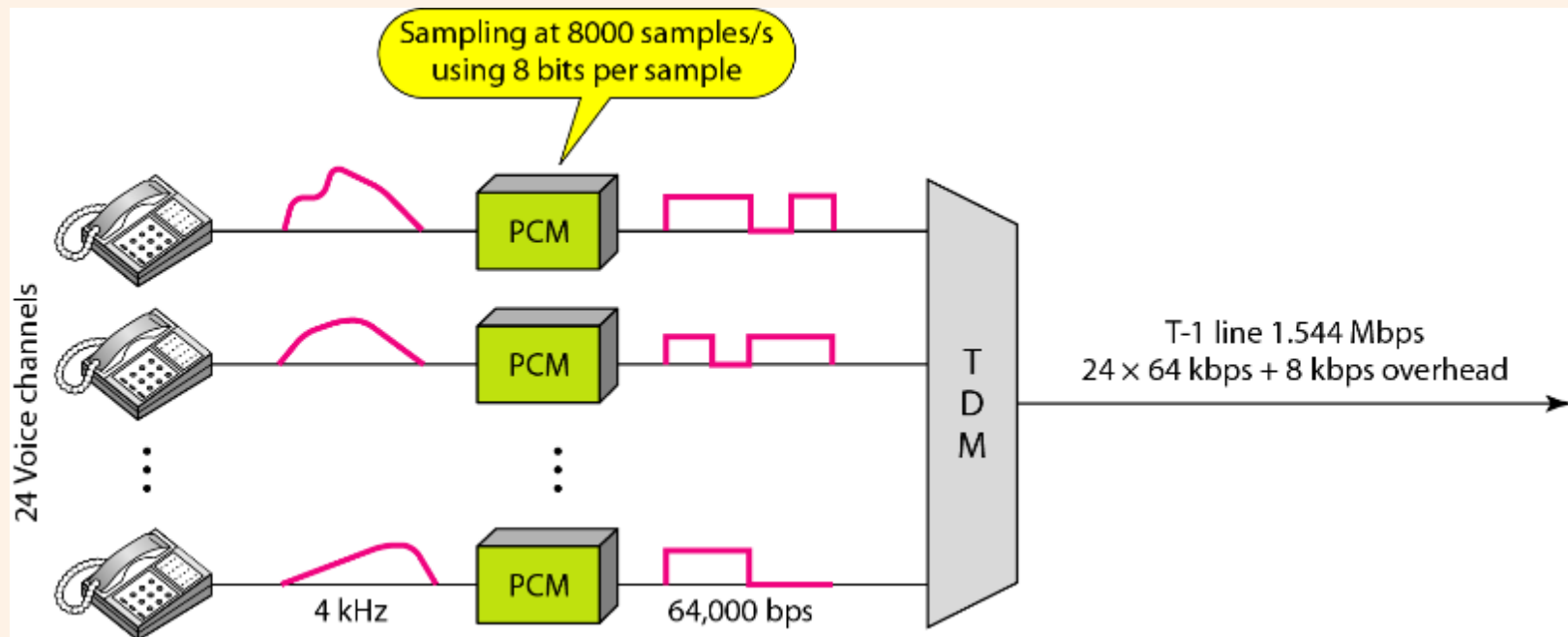


Fig. 26: T-1 line for multiplexing telephone lines.

T-1 frame structure

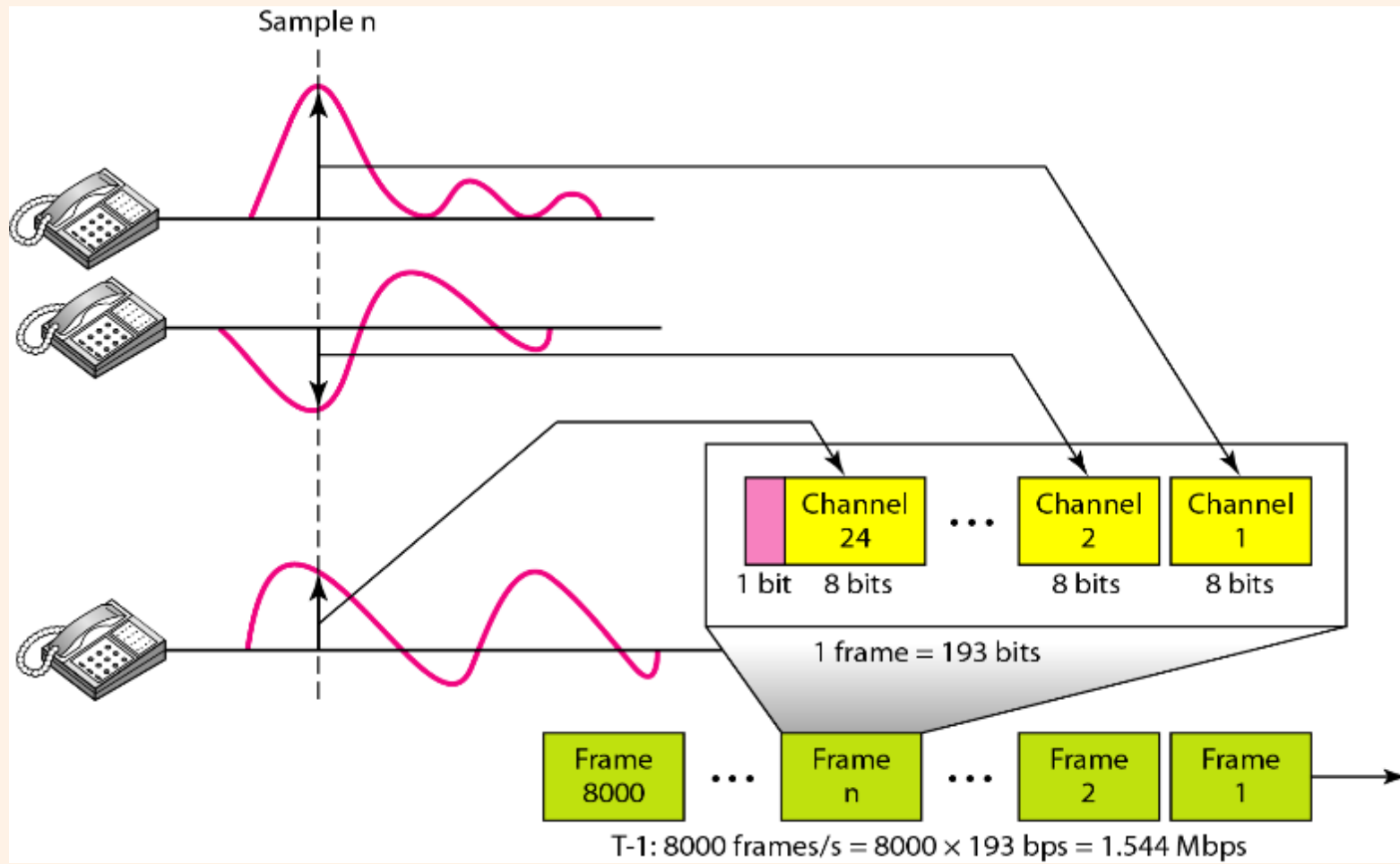


Fig. 27: T-1 Frame Structure.

E line rates

E line rates:

Europeans use a version of T lines called E lines. The two systems are conceptually identical, but their capacities differ.

<i>Line</i>	<i>Rate (Mbps)</i>	<i>Voice Channels</i>
E-1	2.048	30
E-2	8.448	120
E-3	34.368	480
E-4	139.264	1920

Table 2

Statistical Time-Division Multiplexing

In **synchronous TDM**, each input has a **reserved** slot in the **output frame**. This can be **inefficient** if **some input lines** have **no data** to send.

In **statistical time-division multiplexing**,

Slots are **dynamically** allocated to improve **bandwidth efficiency**. Only when an input line has a slot's worth of data to send, then it is given a slot in the output frame.

In **statistical multiplexing**, the **number of slots** in each frame is **less** than the **number of input lines**.

1. The **multiplexer checks** each input line in **round robin** fashion;
2. it **allocates** a slot for an input line if the **line has data to send**;
otherwise,
3. it **skips** the **line** and **checks** the **next line**.

Statistical Time-Division Multiplexing

Addressing:

Figure 28 also shows a major **difference** between slots in **synchronous TDM** and **statistical TDM**. An **output slot** in synchronous TDM is **totally occupied by data**;

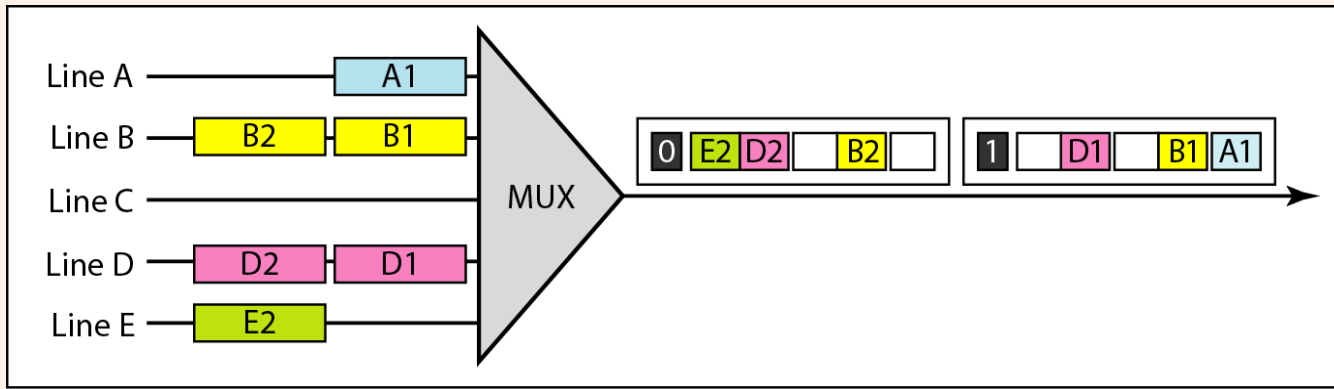
In **statistical TDM**, a slot **needs** to carry **data as well as the address** of the **destination**.

In **synchronous TDM**, there is **no need** for **addressing**; synchronization and pre assigned relationships between the inputs and outputs serve as an address. If the multiplexer and the demultiplexer are synchronized, this is guaranteed.

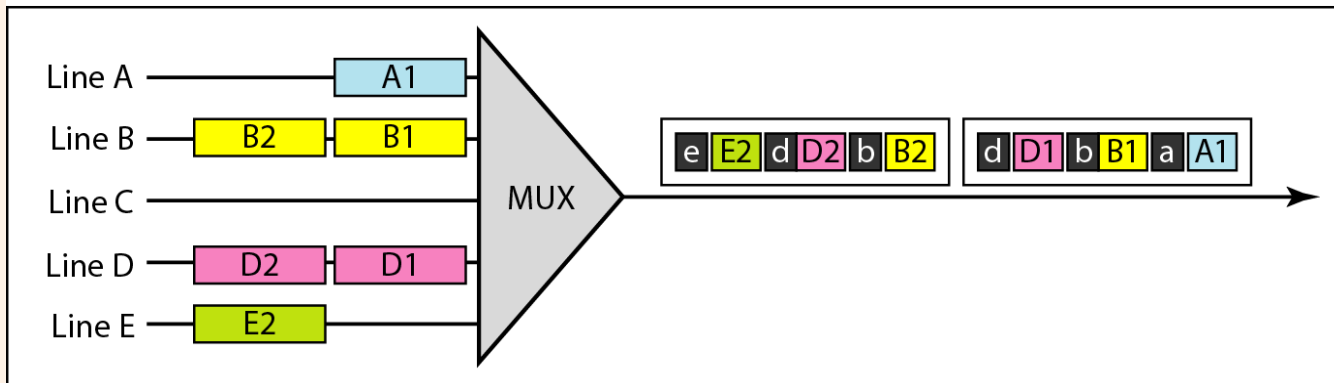
In **statistical multiplexing**, there is **no fixed relationship** between the inputs and outputs because there are **no pre-assigned or reserved slots**.

We **need** to **include** the **address** of the receiver **inside** each slot to show **where** it is to be **delivered**.

TDM slot comparison



a. Synchronous TDM



b. Statistical TDM

Fig. 28: TDM slot comparison.

References

1. Prakash C. Gupta, "Data communications", Prentice Hall India Pvt.
2. William Stallings, "Data and Computer Communications", Pearson
3. Forouzan, B. A. "Data Communication and Networking. Tata McGraw." (2005).